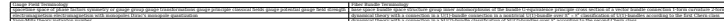


mielke-geometrodynamics: table evidence

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Image																												
mielke-geometrodynamics_TAB_001	75	—	<pre>\begin{tabular}[t]{ l l } \hline Gauge Field Terminology & Fiber Bundle Terminology \\ \hline spacetime space of phase factors symmetry or gauge group gauge transformations gauge principle classical fields gauge potential gauge field strength & base space bundle space structure group inner automorphisms of the bundle G-equivalence principle cross section of a vector bundle connection 1-form curvature 2-form \\ \hline electromagnetism electromagnetism with monopoles Dirac's monopole quantization & dynamical theory with a connection in a U(1)-bundle connection in a nontrivial U(1)-bundle over \mathbb{R}^2 \times \mathrm{S}^2 \\ \hline classification of U(1)-bundles according to the first Chern class \\ \hline Yang-Mills theory instanton number & dynamical theory with a connection in a SU(2)-bundle classification of SU(2)-bundles over \mathrm{S}^4 according to the second Chern class \\ \hline \end{tabular}</pre>		<table><tr><th>Gauge Field Terminology</th><th>Fiber Bundle Terminology</th></tr><tr><td>spacetime</td><td>base space</td></tr><tr><td>space of phase factors</td><td>bundle space</td></tr><tr><td>symmetry or gauge group</td><td>structure group</td></tr><tr><td>gauge transformations</td><td>inner automorphisms of the bundle</td></tr><tr><td>gauge principle</td><td>G-equivalence principle</td></tr><tr><td>classical fields</td><td>cross section of a vector bundle</td></tr><tr><td>gauge potential</td><td>connection 1-form</td></tr><tr><td>gauge field strength</td><td>curvature 2-form</td></tr><tr><td>electromagnetism</td><td>dynamical theory with a connection in a U(1)-bundle</td></tr><tr><td>electromagnetism with monopoles</td><td>connection in a nontrivial U(1)-bundle over $\mathbb{R}^2 \times S^2$</td></tr><tr><td>Dirac's monopole quantization</td><td>classification of U(1)-bundles according to the first Chern class</td></tr><tr><td>Yang–Mills theory</td><td>dynamical theory with a connection in a SU(2)-bundle</td></tr><tr><td>instanton number</td><td>classification of SU(2)-bundles over S^4 according to the second Chern class</td></tr></table>	Gauge Field Terminology	Fiber Bundle Terminology	spacetime	base space	space of phase factors	bundle space	symmetry or gauge group	structure group	gauge transformations	inner automorphisms of the bundle	gauge principle	G-equivalence principle	classical fields	cross section of a vector bundle	gauge potential	connection 1-form	gauge field strength	curvature 2-form	electromagnetism	dynamical theory with a connection in a U(1)-bundle	electromagnetism with monopoles	connection in a nontrivial U(1)-bundle over $\mathbb{R}^2 \times S^2$	Dirac's monopole quantization	classification of U(1)-bundles according to the first Chern class	Yang–Mills theory	dynamical theory with a connection in a SU(2)-bundle	instanton number	classification of SU(2)-bundles over S^4 according to the second Chern class
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mielke-geometrodynamics_TAB_002	102	—	<pre>\begin{tabular}[t]{ l l l l } \hline potential & field strength & Bianchi identity & Noether identity \\ \hline \vartheta^\alpha & T^\alpha = D\vartheta^\alpha & DT^\alpha \equiv R_\gamma{}^\alpha \wedge \vartheta^\gamma & \widehat{D} \Sigma_\alpha \cong (e_\alpha \lrcorner R_\beta{}^\gamma) \wedge \Delta^\beta{}_\gamma - \frac{1}{2} (e_\alpha \lrcorner Q_{\beta\gamma}) \wedge \sigma^{\beta\gamma} \\ \Gamma_\alpha{}^\beta & R_\alpha{}^\beta = d\Gamma_\alpha{}^\beta - \Gamma_\alpha{}^\gamma \wedge \Gamma_\gamma{}^\beta & DR_\alpha{}^\beta \equiv 0 & D\Delta^\alpha{}_\beta + \vartheta^\alpha \wedge \Sigma_\beta \cong \sigma^\alpha{}_\beta \\ \hline \end{tabular}</pre>	(not rendered)	<table><tr><th>potential</th><th>field strength</th><th>Bianchi identity</th><th>Noether identity</th></tr><tr><td>ϑ^α</td><td>$T^\alpha = D\vartheta^\alpha$</td><td>$DT^\alpha \equiv R_\gamma{}^\alpha \wedge \vartheta^\gamma$</td><td>$\widehat{D} \Sigma_\alpha \cong (e_\alpha \lrcorner R_\beta{}^\gamma) \wedge \Delta^\beta{}_\gamma - \frac{1}{2} (e_\alpha \lrcorner Q_{\beta\gamma}) \wedge \sigma^{\beta\gamma}$</td></tr><tr><td>$\Gamma_\alpha{}^\beta$</td><td>$R_\alpha{}^\beta = d\Gamma_\alpha{}^\beta - \Gamma_\alpha{}^\gamma \wedge \Gamma_\gamma{}^\beta$</td><td>$DR_\alpha{}^\beta \equiv 0$</td><td>$D\Delta^\alpha{}_\beta + \vartheta^\alpha \wedge \Sigma_\beta \cong \sigma^\alpha{}_\beta$</td></tr></table>	potential	field strength	Bianchi identity	Noether identity	ϑ^α	$T^\alpha = D\vartheta^\alpha$	$DT^\alpha \equiv R_\gamma{}^\alpha \wedge \vartheta^\gamma$	$\widehat{D} \Sigma_\alpha \cong (e_\alpha \lrcorner R_\beta{}^\gamma) \wedge \Delta^\beta{}_\gamma - \frac{1}{2} (e_\alpha \lrcorner Q_{\beta\gamma}) \wedge \sigma^{\beta\gamma}$	$\Gamma_\alpha{}^\beta$	$R_\alpha{}^\beta = d\Gamma_\alpha{}^\beta - \Gamma_\alpha{}^\gamma \wedge \Gamma_\gamma{}^\beta$	$DR_\alpha{}^\beta \equiv 0$	$D\Delta^\alpha{}_\beta + \vartheta^\alpha \wedge \Sigma_\beta \cong \sigma^\alpha{}_\beta$																
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mielke-geometrodynamics_ TAB_003	210	—	$\begin{tabular}{t}{ l l l l l l }\hline & \text{Objects} & & \text{(p)-form} & \text{Components} & \text{(\mathrm{n}=4)} & 3 & 2 \\ \hline & \vartheta^\alpha & \text{Vector} & 1 & n^2 & 16 & 9 & 1 \\ & \Gamma_\alpha^\star & \text{Vector} & 1 & n^2 & 16 & 9 & 1 \\ & T^\alpha & \text{Vector} & 2 & n^2(n-1)/2 & 24 & 9 & 2 \\ & R^{\alpha\beta} & \text{Bivector} & 2 & n^2(n-1)^2/4 & 36 & 9 & 1 \\ & \Sigma_\alpha & \text{Vector} & n-1 & n^2 & 16 & 9 & 4 \\ & \tau_{\alpha\beta} & \text{Bivector} & n-1 & n^2(n-1)/2 & 24 & 9 & 2 \\ & \eta_\alpha & \text{Vector} & n-1 & n^2 & 16 & 9 & 4 \\ \hline \end{tabular}$	<table><tr><td></td><td>Objects</td><td>p-form</td><td>Components</td><td>n = 4</td><td>3</td><td>2</td></tr><tr><td>ϑ^α</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>$\Gamma_\alpha^\star$</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>T^α</td><td>Vector</td><td>2</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>$R^{\alpha\beta}$</td><td>Bivector</td><td>2</td><td>$n^2(n-1)^2/4$</td><td>36</td><td>9</td><td>1</td></tr><tr><td>Σ_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr><tr><td>$\tau_{\alpha\beta}$</td><td>Bivector</td><td>$n-1$</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>η_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr></table>		Objects	p -form	Components	n = 4	3	2	ϑ^α	Vector	1	n^2	16	9	1	$\Gamma_\alpha^\star$	Vector	1	n^2	16	9	1	T^α	Vector	2	$n^2(n-1)/2$	24	9	2	$R^{\alpha\beta}$	Bivector	2	$n^2(n-1)^2/4$	36	9	1	Σ_α	Vector	$n-1$	n^2	16	9	4	$\tau_{\alpha\beta}$	Bivector	$n-1$	$n^2(n-1)/2$	24	9	2	η_α	Vector	$n-1$	n^2	16	9	4	<table><tr><td></td><td>Objects</td><td>p-form</td><td>Components</td><td>n = 4</td><td>3</td><td>2</td></tr><tr><td>ϑ^α</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>$\Gamma_\alpha^\star$</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>T^α</td><td>Vector</td><td>2</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>$R^{\alpha\beta}$</td><td>Bivector</td><td>2</td><td>$n^2(n-1)^2/4$</td><td>36</td><td>9</td><td>1</td></tr><tr><td>Σ_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr><tr><td>$\tau_{\alpha\beta}$</td><td>Bivector</td><td>$n-1$</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>η_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr></table>		Objects	p -form	Components	n = 4	3	2	ϑ^α	Vector	1	n^2	16	9	1	$\Gamma_\alpha^\star$	Vector	1	n^2	16	9	1	T^α	Vector	2	$n^2(n-1)/2$	24	9	2	$R^{\alpha\beta}$	Bivector	2	$n^2(n-1)^2/4$	36	9	1	Σ_α	Vector	$n-1$	n^2	16	9	4	$\tau_{\alpha\beta}$	Bivector	$n-1$	$n^2(n-1)/2$	24	9	2	η_α	Vector	$n-1$	n^2	16	9	4
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mielke-geometrodynamics_ TAB_004	275	—	$\begin{tabular}{t}{ l l l }\hline & \text{Spin} & \text{Gravitational} & \text{YM anomaly} \\ \hline & 1/2 & 1 & 1 \\ & 3/2 & -21 & 3 \\ \hline \end{tabular}$	<table><tr><td>Spin</td><td>Gravitational</td><td>YM anomaly</td></tr><tr><td>1/2</td><td>1</td><td>1</td></tr><tr><td>3/2</td><td>-21</td><td>3</td></tr></table>	Spin	Gravitational	YM anomaly	1/2	1	1	3/2	-21	3	<table><tr><td>Spin</td><td>Gravitational</td><td>YM anomaly</td></tr><tr><td>1/2</td><td>1</td><td>1</td></tr><tr><td>3/2</td><td>-21</td><td>3</td></tr></table>	Spin	Gravitational	YM anomaly	1/2	1	1	3/2	-21	3																																																																																														
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